



# Follow the Steps

Run the algorithm to make a pattern

Name: \_\_\_\_\_

Date: \_\_\_\_\_

★ I can follow the steps of an algorithm to make a pattern.

## Run the algorithm

Follow the steps in order, one at a time, to fill the boxes. Each step tells you which shape to draw in the next box. Running the steps makes a pattern.

## The algorithm

1. box 1 → draw ●

2. box 2 → draw ■

3. box 3 → draw ●

4. box 4 → draw ■

5. box 5 → draw ●

6. box 6 → draw ■

## Draw here — boxes 1 to 6

|  |  |  |  |  |  |
|--|--|--|--|--|--|
|  |  |  |  |  |  |
|--|--|--|--|--|--|

### 1 Follow the steps

Do each step in order and draw ● or ■ in the boxes above. Read box 1 first, then box 2, and so on.

## 2 Read your result

What shape is in box 6? Draw it or write its name.

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## 3 You try — keep the pattern

What is the pattern called (● ■ ● ■ ...)? If there were a box 7, what shape would go there? Tell how you know.

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### Big idea

Running an algorithm means doing each step in order. The same steps always make the same result — here, a circle-square pattern.



# Follow the Steps

Quick facilitation notes & answer key

## What this worksheet teaches

Students follow a 6-step set of instructions in order to fill a row of boxes, making a circle-square pattern, then read and continue it. Following steps in order to build a result is running a plan. No coding experience needed.

## Before you start (no computer needed)

No computer — paper and a pencil. You read each prompt aloud. Optional: cut the steps into cards so students can physically slide them into order before writing.

## How to lead it (about 20 minutes)

1. Show them (you do). Read box 1, draw its shape; read box 2, draw its shape — one box at a time, in order.
2. Do one together. Fill the first few boxes together (circle, square, circle...).
3. Let them try. Students finish the row, then read and continue the pattern.

## Answer key (these are the verified correct answers)

Exercise 1 — boxes 1–6: circle, square, circle, square, circle, square. Exercise 2 — box 6 is a square.

Exercise 3 — the pattern is circle-square (an AB pattern); box 7 would be a circle, because the pattern repeats and box 6 ended on a square.

Why it works: each step runs in order and changes only its own box, so the row always comes out the same.

## If students get stuck — and ways to adjust

Watch for: skipping a box or doing them out of order. Read the boxes strictly 1, 2, 3...

Make it easier: point to each box as you read its step.

Make it harder: ask what box 8 would be and how they know.

## Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students put step-by-step instructions in the right order to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read a set of steps, change or reorder one, and explain what the change did.

## You'll know it worked when

The child fills the row in order and continues the pattern correctly.